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Summary

Earth has numerous habitats with a wide range of living organisms inhabiting it. Living organisms show a variety of unique characters different from non-living matter. Classification is the process by which anything is grouped in a convenient category based on some easily observable characters.

Taxonomy is the science of arrangement of living organisms. R. H. Whittaker proposed the five kingdom classification. Three domain classification was proposed by Carl Woese and his co-workers.

The taxonomical hierarchy includes seven categories namely kingdom, phylum, class, order, family, genus and species. The process of assigning scientific names to animal or taxonomic group is called nomenclature. Each scientific name has two components, generic name and a specific epithet. The important component of the taxonomical tools are field visits, survey, identification, classification, preservation and documentation. Molecular taxonomical tools are more accurate, authentic and significant for taxonimical classification.

Glossary

Cladogram – A branching diagram showing the relationship between a number of species.

Phylogeny – Relationships among various biological species based upon similarities and differences in their physical or genetic characteristics.

Phylogenetic tree – A phylogenetic tree or evolutionary tree is a branching diagram or "tree" showing the inferred evolutionary relationships upon similarities and differences in their physical or genetic characteristics.

Shared character – A shared character is one that two lineages have in common

Derived character – Derived character is one that evolved in the lineage leading up to a clade.

Threatened species – Species which are vulnerable to endangerment in the near future.

Evaluation

1. A living organism is differentiated from non-living structure based on
   a. Reproduction
   b. Growth
   c. Metabolism
   d. Movement

2. A group of organisms having similar traits of a rank is
   a. Species
   b. Taxon
   c. Genus
   d. Family